

A Practical Engineering Blueprint for Scaling Geospatial Features to All-India at CatBoost Inference Speeds

National OSM Data Acquisition and Extraction

The foundational step in scaling the geospatial feature engineering pipeline from a regional pilot to a national deployment is the acquisition of OpenStreetMap (OSM) data for all of India. The initial research question presented a choice between using Overpass API and a bulk PBF extract. However, given the constraints of a nationwide query, the user's directive clarifies that Overpass API is fundamentally unsuitable for this scope, regardless of optimization priorities ¹ ¹¹⁰. Its documented fair-use policies explicitly discourage large-area queries, and its architecture is not designed to handle the computational load of processing an entire country's worth of data in a single request, making it prone to timeouts and rate limiting ¹⁰⁹. Therefore, the only viable and robust approach is to procure a bulk PBF (Protocolbuffer Binary Format) extract and perform local processing. The primary source for such extracts is Geofabrik, which provides regularly updated continent and country-sized files ¹² ¹⁵. For a single-machine setup typical of a hackathon environment, a country-specific extract is far more manageable than a larger regional one. The latest available extract for India has a file size of approximately 1.6 GB ¹⁰. This size is reasonable for download times over a stable internet connection and for subsequent processing without requiring specialized hardware.

Once the PBF file is acquired, the next critical task is to efficiently filter it to extract only the necessary tag categories: `landuse=industrial`, `landuse=quarry`, `man_made=mineshaft`, and `man_made=adit`. Processing the entire ~1.6 GB file would be inefficient. The recommended tool for this task is `osmium-tool`, a high-performance library specifically designed for reading and writing OSM data in PBF format ⁵ ²⁴⁶. Unlike libraries that parse XML, `osmium` operates directly on the binary data structure, offering significant speed advantages. An effective command-line workflow involves using `osmium tags-filter` to isolate the desired ways and relations based on their tags. A command such as `osmium tags-filter -o india_industrial_quarry.osm.pbf india-latest.osm.pbf way/rel`

landuse=industrial landuse=quarry man_made=mineshaft
man_made=adit can create a new, smaller PBF file containing only the relevant elements. This avoids loading the entire dataset into memory, making the operation feasible on a standard machine. While no precise benchmark was found for filtering a single Indian state, `osmium-tool`'s performance on continent-sized extracts demonstrates its capability; processing the entire Asia region (~15.1 GB) is possible in under an hour on modern hardware ^{11 246}. It is therefore reasonable to estimate that filtering the smaller 1.6 GB India file would take well under one hour, fitting comfortably within the project's timeline.

Finally, it is crucial to verify the continued validity of the selected OSM tags. The tagging conventions within the OSM community can evolve over time, and tags may be deprecated or superseded by new ones. The tags specified in the research goal—`landuse=industrial`, `landuse=quarry`, `man_made=mineshaft`, and `man_made=adit`—are widely recognized and appear in various academic and commercial applications related to industrial and mining activity ^{4 81}. However, a quick manual verification through the OSM website or documentation is a prudent final check. For instance, searching for "mineshaft" on the OSM wiki confirms its current usage. Should any of these tags be found to be deprecated, `osmium` offers capabilities to remap tags, though this adds a layer of complexity that should be avoided if possible. Given the prevalence of these tags, they are expected to remain current. By adopting the bulk PBF extraction method via Geofabrik and using `osmium-tool` for filtering, the team can reliably acquire the necessary OSM data within the hackathon's tight schedule.

Parameter	Specification	Justification
Primary Data Source	Geofabrik Download Server ¹²	Provides reliable, regularly updated continent/country PBF extracts.
Target File	India (latest) ¹⁰	Country-specific extract is manageable for single-machine processing.
Estimated File Size	~1.6 GB ¹⁰	Reasonable download time and storage requirement for a hackathon setup.
Recommended Tool	<code>osmium-tool</code> ^{5 246}	High-performance binary format processor, significantly faster than XML parsers.
Estimated Processing Time	< 1 hour	Based on extrapolation from benchmarks for continent-sized extracts ^{11 246} .
Target Tags	<code>landuse=industrial</code> , <code>landuse=quarry</code> , <code>man_made=mineshaft</code> , <code>man_made=adit</code>	Widely used tags for industrial, quarry, and mine infrastructure ^{4 81} .

WRI GPPD Coverage and Fallback Dataset Assessment

The utility of the `dist_wri_facility` feature hinges entirely on the completeness of the World Resources Institute's Global Power Plant Database (GPPD) for gas, oil, and coal facilities within India. Before committing to integrating this dataset, its national coverage must be quantified to determine if it provides a statistically meaningful signal. The analysis should proceed programmatically to ensure objectivity. The first step is to download the latest public release of the WRI GPPD, which serves as a comprehensive global dataset for power plants ^{18 22}. Subsequently, the dataset must be filtered to isolate records corresponding to India and relevant fuel types. This involves two stages of filtering: first, selecting all entries where the country code or name is 'India', and second, further narrowing the selection to facilities where the `primary_fuel` attribute falls within the set ['Gas', 'Oil', 'Coal']. The resulting count will dictate the strategic path forward.

The interpretation of this count is critical and directly informs whether the feature is valuable or merely dead weight. If the WRI GPPD yields a substantial number of qualifying facilities nationally (e.g., hundreds or thousands), the feature remains highly valuable. Even if the distribution is uneven, the continuous distance metric will effectively capture the strong signals in high-density areas like Gujarat and Maharashtra while still providing a weak but potentially useful signal elsewhere. In this scenario, proceeding with WRI GPPD is the most efficient course of action, as it avoids the significant engineering overhead of sourcing, parsing, and integrating a secondary dataset. The user's guidance rightly emphasizes avoiding this cost unless absolutely necessary, especially under a tight hackathon timeline ¹⁹⁵¹⁹⁹. The primary limitation would then be acknowledged transparently, a practice already established for other aspects of the project.

However, if the analysis reveals sparse coverage, the strategy must pivot. If the count is low (e.g., dozens of facilities), the feature's utility becomes questionable. It might act as a powerful positive indicator when a hotspot is located near one of these rare points, but provide almost no information for the vast majority of detections. This represents a moderate limitation that could still be included in the model, with the understanding of its restricted applicability. The decision to move to a fallback dataset should only be triggered if the coverage is truly unusable, meaning the count is extremely low (e.g., zero or a handful). At this point, the engineering cost of adding a new data source becomes justifiable to salvage the feature's purpose. The user has provided a clear recommendation for a fallback: the data from the Petroleum and Natural Gas Regulatory Board (PNGRB) ⁹⁶. PNGRB is the better choice because it is specific to oil and gas

infrastructure, aligning directly with the "gas flare" classification goal of the model. Integrating this data, however, carries a non-trivial risk. Government datasets are often distributed in messy formats like PDF reports or HTML web pages, requiring significant effort for automated parsing and schema alignment ³⁴ ¹⁹⁷. The Central Electricity Authority (CEA) of India provides general power plant lists, but this data is more redundant with WRI's existing scope and less targeted towards the specific oil/gas use case, making it a less ideal fallback option ⁹⁷. Therefore, the recommended path is to first quantify WRI's coverage; only if it proves to be genuinely insufficient should the team undertake the higher-risk task of integrating PNGRB data.

Spatial Join Methodology for Large-Scale Proximity Analysis

Selecting the appropriate spatial join methodology is paramount for achieving acceptable performance when calculating proximity features for an all-India H3 res-8 grid. The scale of the problem dictates that naive, brute-force methods are untenable. The first step in assessing feasibility is to establish realistic row counts. At H3 resolution 8, each hexagon has an approximate area of 0.74 km² ². With India's total land area being around 3.1 million km², the number of H3 cells covering the country is approximately 4.2 million. For the OSM polygon data, the volume varies by tag. For common features like farmland, the count could easily reach into the millions, whereas rarer features like mineshafts might number in the thousands. The nearest-neighbor join will need to be performed for each of these polygon categories against the 4.2 million H3 cells.

Given these scales, the choice of algorithm is critical. The most practical and recommended starting point is `geopandas.sjoin_nearest` ¹⁰⁸. This function is part of the high-level GeoPandas library, which simplifies geospatial workflows in Python ²¹⁸. Crucially, it leverages underlying spatial indexes, such as an STRtree, to dramatically accelerate the search for nearest neighbors ¹⁷⁵ ²⁵². Benchmarks have shown that even complex spatial joins involving hundreds of millions of points can be executed in minutes on a capable machine when optimized with an index ¹¹⁵. For a hackathon setting, its combination of ease of use, good average-case performance, and built-in optimizations makes it the ideal candidate. The key to its efficiency lies in creating a spatial index on the OSM polygon GeoDataFrame before performing the join.

An alternative approach involves using lower-level libraries like `scipy.spatial.cKDTree` or `BallTree` ⁵⁸. These structures are highly optimized for point-based nearest-neighbor searches and offer flexibility in defining distance metrics ²¹⁷. However, they are not designed to work directly with polygons. To use them, each polygon must first be reduced to a representative point, typically its centroid ¹⁸⁶. This introduces a fundamental approximation error, as the distance from an H3 cell centroid to a polygon's centroid is not equivalent to the true minimum distance from the cell to the polygon boundary ¹⁵⁵²²¹. While this might be sufficient for some analyses, it compromises the accuracy of the feature, which is undesirable for a machine learning model. Furthermore, while `KDTree` and `BallTree` are fast for point data, the additional step of coordinate transformation and the potential for increased error make them a less robust choice compared to the native polygon-aware handling of `geopandas`.

R-trees represent another class of spatial indexes, which is the underlying technology used by `geopandas` for its spatial operations ⁵⁶. Building an R-tree index on the OSM polygons before the join is a crucial optimization that should be performed regardless of the chosen high-level function. The efficiency of an R-tree depends on the characteristics of the geometries; it performs best when there are few overlapping bounding boxes, which is often the case for spatial data ¹¹¹. The takeaway is that the focus should be on leveraging a spatial index, and `geopandas.sjoin_nearest` provides the most straightforward and accurate implementation for this specific point-to-polygon join task. The performance of this method is likely to hold up well at the estimated scale of 4.2 million H3 cells against a few hundred thousand polygons, making it the optimal choice for the hackathon timeline.

Method	Pros	Cons	Recommendation
<code>geopandas.sjoin_nearest</code>	High-level API, easy to use; leverages spatial indexes (STRtree); handles polygons directly; good average performance ¹⁰⁸¹⁷⁵²⁵² .	Performance depends on underlying GEOS/Spatial Index implementation.	Recommended. Balances development speed, accuracy, and performance for this use case.
<code>scipy KDTree / BallTree</code>	Very fast for point-based searches; flexible distance metrics ^{58 217} .	Requires approximating polygons with centroids, introducing error; not natively designed for polygons ¹⁸⁶ .	Not recommended due to inherent approximation error.
Manual R-tree Implementation	Can be highly optimized.	More complex to implement correctly; <code>geopandas</code> already uses this internally, so building it manually provides no advantage.	Not necessary. Use <code>geopandas</code> which abstracts this away.

Nationwide Distance Calculation and CRS Strategy

Accurately calculating distances across the entirety of India presents a significant challenge due to its geographical extent, which spans multiple UTM (Universal Transverse Mercator) zones ²⁷³. Using planar coordinates from a single UTM zone for calculations anywhere else on the map would introduce substantial distortion and inaccurate results. The user's clarification highlights that the previous work in Gujarat and Maharashtra relied on a single UTM zone, a method that cannot be naively scaled nationally. Therefore, a standardized and globally consistent methodology is required. Three primary approaches exist: using a single custom equal-area projection, performing direct geodesic calculations, or reprojecting data on a per-state basis.

Reprojecting data on a per-state basis is computationally prohibitive and introduces significant complexity. For every distance calculation involving an H3 cell, the system would need to determine the cell's UTM zone, reproject both the cell and the target polygon into that zone's coordinate system, calculate the distance, and then discard the result. This constant transformation overhead for millions of calculations makes it an impractical solution for batch processing, let alone for real-time inference.

Using a single custom projection, such as a Lambert Azimuthal Equal-Area centered on India's geographic center, is a viable compromise ²⁶ ²⁷. Such projections are specifically designed to minimize distortion across a continental-scale area, preserving either area or distance from the central point ²⁸ ³⁰. While this approach is more accurate than using a single UTM zone, it still relies on an approximation. The accuracy of distance measurements decreases as you move away from the projection's center. For a machine learning model, maintaining consistency in how distance is measured is crucial, and any projection-based method introduces a dependency on the chosen parameters and potential artifacts.

The third and most robust approach is to use direct geodesic calculations. This method computes the shortest path between two points on an ellipsoid, which is a much more accurate representation of the Earth's surface than a flat plane ²⁶⁸²⁶⁹. The `pyproj` library, which is part of the standard geospatial Python stack and is bundled with GeoPandas, provides a `Geod` class for this purpose ³⁸ ⁷⁶. Functions within this class can compute geodesic distances, lengths, and areas directly from latitude and longitude coordinates ³⁵ ³⁶. While geodesic computations are mathematically more intensive than simple Euclidean distance calculations in a projected CRS, the performance difference is often negligible for batch processing tasks involving millions of points, especially since `pyproj` is highly optimized ⁹⁴. For the specific use case of geospatial machine learning,

where model training requires consistent and accurate feature representation, the benefits of geodesic distance far outweigh the minor computational cost. It eliminates the need to manage multiple Coordinate Reference Systems (CRS) and ensures that a unit of distance is consistent everywhere in India. This simplicity, combined with its high accuracy, makes geodesic calculation the standard and recommended practice for this type of nationwide analysis.

Precompute-once Strategy Feasibility and Operational Recommendation

The final engineering decision concerns the operational architecture for generating the proximity features: a precompute-once strategy versus an on-demand computation approach. The user's clarification provides a decisive framework for this decision, prioritizing demo reliability and engineering simplicity over theoretical considerations of latency and storage. In a live dashboard demonstration, predictability and deterministic behavior are paramount. Any runtime dependency that can fail, slow down, or behave unexpectedly during a judge's evaluation is a significant liability. This context immediately favors the precompute-once strategy.

The precompute-once approach involves executing a comprehensive batch job to calculate all proximity features (e.g., `dist_industrial`, `dist_quarry`, etc.) for every cell in the fixed India-wide H3 res-8 grid once. The results are stored in a lookup table. During inference, when a FIRMS hotspot detection arrives, the system simply looks up the precomputed feature values for the H3 cell the detection falls into. This architecture is highly reliable because the heavy computational work is done offline and can be fully tested beforehand. The inference pipeline is reduced to a simple database query, removing a major potential failure mode from the live demo.

Conversely, an on-demand strategy keeps the raw OSM/WRI data and a spatial index in memory. For each incoming detection, the system calculates the proximity features on the fly. While both approaches would likely be "fast enough" for a demo's request volume, the on-demand method introduces a runtime dependency on the spatial indexing system. This system could experience connectivity issues, suffer from performance degradation under load, or return unexpected results, all of which could disrupt the demonstration 77. Given that demo reliability is the primary criterion, the risk associated with the on-demand approach is unacceptable.

The remaining question is whether the precomputation batch job itself is feasible within the 3-week hackathon timeline. The feasibility can be broken down into its constituent parts. First, acquiring and filtering the OSM data, as detailed previously, is estimated to take less than an hour. Second, the most computationally intensive step is the series of spatial joins. Performing four separate nearest-neighbor joins for approximately 4.2 million H3 cells against several collections of OSM polygons (e.g., industrial, quarry, mineshaft, farmland) is a demanding task. While a precise benchmark for this exact workload is unavailable, estimates based on similar large-scale geospatial operations suggest this step could take several hours to a day on a capable machine. This is a hard deadline for the hackathon but is achievable with focused optimization. Third, the final step of calculating the geodesic distances for all these pairs using `pyproj.Geod` is also intensive but is likely to complete relatively quickly due to the library's optimization, perhaps within minutes to an hour. The total estimated runtime for the precomputation job is therefore in the range of several hours to a day, which fits within the project's overall schedule.

The storage cost for the precomputed feature table is trivial. The output would consist of the H3 cell ID and a small number of continuous float columns (one for each distance feature) and a few binary flag columns. With approximately 4.2 million rows, this dataset would occupy a very small amount of disk space, likely well under a gigabyte. Storage is not a constraint. Therefore, the precompute-once strategy is unequivocally the correct choice. It maximizes demo reliability and simplifies the architecture, and the computational and storage costs of the batch job are manageable within the given timeframe.

Reference

1. Finding nearest point in other GeoDataFrame using ... <https://gis.stackexchange.com/questions/222315/finding-nearest-point-in-other-geodataframe-using-geopandas>
2. Working with FME and Uber's H3 Data <https://support.safe.com/hc/en-us/articles/25407601950093-Working-with-FME-and-Uber-s-H3-Data>
3. The Engineer's Field Guide to Geospatial Analytics, Route ... <https://medium.com/@brian-curry-research/the-engineers-field-guide-to-geospatial-analytics-route-optimization-and-gis-963478bbfb79>

4. 75+ Geospatial Python and Spatial Data Science ... <https://forrest.nyc/75-geospatial-python-and-spatial-data-science-resources-and-guides/>
5. OPENSTREETMAP ELEMENT VECTORISATION - A TOOL ... <https://isprs-archives.copernicus.org/articles/XLVIII-4-W1-2022/395/2022/isprs-archives-XLVIII-4-W1-2022-395-2022.pdf>
6. Enhancing geospatial retail analysis by integrating ... <https://www.sciencedirect.com/science/article/pii/S0198971523001217>
7. Sylvain Lobry on Visual Question Answering on Remote ... https://www.linkedin.com/posts/vahid-akbari-b431131a_vqa-openstreetmap-sar-activity-7398345663010451456-op_1
8. Climate Change AI Workshop Papers <https://www.climatechange.ai/papers>
9. ISRS 2026 Technical Program https://www.rssj.or.jp/isrs2026/program_April_14.html
10. Download OpenStreetMap for India <https://download.geofabrik.de/asia/india.html>
11. Download OpenStreetMap for Asia | Geofabrik Download Server <https://download.geofabrik.de/asia.html>
12. GEOFABRIK // Downloads <https://www.geofabrik.de/data/download.html>
13. Download OpenStreetMap for Central Zone <https://download.geofabrik.de/asia/india/central-zone.html>
14. North-Eastern Zone - OpenStreetMap Data Extracts - Geofabrik <https://download.geofabrik.de/asia/india/north-eastern-zone.html>
15. OpenStreetMap Data Extracts - Geofabrik <https://download.geofabrik.de/>
16. These 10 Countries Are Phasing Out Coal the Fastest <https://www.wri.org/insights/countries-phasing-out-coal-power-fastest>
17. Redefining America's 'Energy Communities' Can Boost ... <https://www.wri.org/insights/defining-americas-energy-communities>
18. These Charts Explain Greenhouse Gas Emissions by Sector <https://www.wri.org/insights/4-charts-explain-greenhouse-gas-emissions-countries-and-sectors>
19. The History of Carbon Dioxide Emissions <https://www.wri.org/insights/history-carbon-dioxide-emissions>
20. 7 Ocean-Based Solutions to Fight Climate Change <https://www.wri.org/insights/ocean-based-climate-change-solutions>
21. World's Top Emitters Interactive Chart I ... <https://www.wri.org/insights/interactive-chart-shows-changes-worlds-top-10-emitters>
22. Biomass for Carbon Removal, Explained <https://www.wri.org/insights/sustainable-biomass-carbon-removal>

23. Reduce Greenhouse Gas Emissions from Agricultural ... <https://research.wri.org/wrr-food/course/reduce-greenhouse-gas-emissions-agricultural-production-synthesis>
24. CARBON - World Resources Report http://pdf.wri.org/leveling_the_carbon_playing_field.pdf
25. Hungary: Converting the Mátra Power Plant into a ... <https://www.wri.org/snapshots/hungary-converting-matra-power-plant-diversified-hub-green-energy-and-industry>
26. Lambert azimuthal equal-area | ArcGIS Pro documentation <https://doc.esri.com/en/arcgis-pro/latest/help/mapping/properties/lambert-azimuthal-equal-area.html>
27. Lambert azimuthal equal-area projection https://en.wikipedia.org/wiki/Lambert_azimuthal_equal-area_projection
28. Lambert Azimuthal Equal Area Projection https://surferhelp.goldensoftware.com/projections/Lambert_Azimuthal_Equal_Area_Projection.htm
29. Lambert Azimuthal Equal Area https://gdal.org/proj_list/lambert_azimuthal_equal_area.html
30. North Pole Lambert Azimuthal Equal Area (Europe) <https://epsg.io/17299>
31. A schematic of the Lambert azimuthal equal-area projection. https://www.researchgate.net/figure/A-schematic-of-the-Lambert-azimuthal-equal-area-projection_fig4_359212694
32. Experience <https://experience.arcgis.com/experience/d60f2a74359c4d5abb00631e5ede9689>
33. Various CRS projections for use with {sf} plots in {ggplot2} https://aditya-dahiya.github.io/visage/geocomputation/crs_projections.html
34. <https://gis.nwic.in/portal/home/item.html?id=6d1af...> <https://gis.nwic.in/portal/home/item.html?id=6d1af6bece3e4f9782dc2f7da37e1142>
35. Geod - pyproj 3.7.2 documentation <https://pyproj4.github.io/pyproj/stable/api/geod.html>
36. Getting Started - pyproj 3.7.2 documentation <https://pyproj4.github.io/pyproj/stable/examples.html>
37. Geod — pyproj 3.0.1 documentation <https://pyproj4.github.io/pyproj/3.0.1/api/geod.html>
38. Geod — pyproj 2.3.1 documentation <https://pyproj4.github.io/pyproj/v2.3.1rel/api/geod.html>
39. Proj - pyproj 3.7.2 documentation <https://pyproj4.github.io/pyproj/stable/api/proj.html>
40. CRS - pyproj 3.7.3.dev0 documentation <https://pyproj4.github.io/pyproj/dev/api/crs/crs.html>

41. Gotchas/FAQ - pyproj 3.7.2 documentation <https://pyproj4.github.io/pyproj/stable/gotchas.html>
42. 12. Working with CRS and Projections in Python https://hamedalemo.github.io/advanced-geo-python/lectures/crs_python.html
43. Map projections https://autogis-site.readthedocs.io/en/2018_/notebooks/L2/projections.html
44. 6 Reprojecting geographic data - Geocomputation with Python <https://py.geocompx.org/06-reproj>
45. A comprehensive GeoAI review: Progress, Challenges and ... <https://arxiv.org/html/2412.11643v1>
46. sacridini/Awesome-Geospatial: Long list of ... <https://github.com/sacridini/Awesome-Geospatial>
47. My Favorite Cartographers of All Time | by Atmika Pai <https://medium.com/between-the-graticules/notes-on-esri-cartography-mooc-f0c4040f1d01>
48. Statistical Inferences and Predictions for Areal Data ... <https://arxiv.org/html/2208.07900v3>
49. A Geospatially Aware Deep Learning Framework for Crop ... <https://arxiv.org/html/2607.18779v2>
50. A comprehensive GeoAI review: Progress, Challenges and ... <https://arxiv.org/html/2412.11643>
51. Unsupervised Detection of Entry and Exit Regions from ... <https://arxiv.org/html/2607.10949>
52. Techniques for deep learning with satellite & aerial imagery <https://github.com/satellite-image-deep-learning/techniques>
53. Treatment Geometry and Causal Identification with Earth ... <https://arxiv.org/pdf/2607.19908>
54. A Practical Recipe for Field Boundary Segmentation at Scale <https://arxiv.org/html/2603.27101v1>
55. Finding all possible adjacent/closest geometrical points ... <https://gis.stackexchange.com/questions/368478/finding-all-possible-adjacent-closest-geometrical-points-from-a-given-point>
56. Python Spatial Indexing: Accelerating Geographic Queries <https://medium.com/@stacyfuende/python-spatial-indexing-accelerating-geographic-queries-f9bde82b8fe3>
57. K-nearest points from two dataframes with GeoPandas <https://stackoverflow.com/questions/62198199/k-nearest-points-from-two-dataframes-with-geopandas>
58. Ball Tree and KD Tree Algorithms <https://www.geeksforgeeks.org/machine-learning/ball-tree-and-kd-tree-algorithms/>

59. OSM Data Fundamentals & Architecture <https://www.osm-data-processing.org/osm-data-fundamentals-architecture/>
60. New OSM file format: 30% smaller than PBF, 5x faster to ... <https://community.openstreetmap.org/t/new-osm-file-format-30-smaller-than-pbf-5x-faster-to-import/137151>
61. Analysis of OSM building data completeness using new data <https://heigit.org/analysis-of-osm-building-data-completeness-using-new-data/>
62. Geospatial analysis using Python course with industry ... <https://www.facebook.com/groups/1365905540464749/posts/2518960268492598/>
63. Enhancing Spatial Accuracy of OpenStreetMap Data <https://www.iieta.org/journals/mmep/paper/10.18280/mmep.100630>
64. Application of Geospatial Techniques and the MCDM Method ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC9924209/>
65. Exploring the Potential of OpenStreetMap Data in Regional ... <https://www.mdpi.com/2072-4292/16/2/239>
66. Feature selection with distance correlation | Phys. Rev. D <https://link.aps.org/doi/10.1103/PhysRevD.109.054009>
67. AI techniques for feature extraction, simplification and ... <https://ertjournal.org/index.php/pub/article/view/597>
68. AI Techniques for Feature Extraction, Simplification and ... https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7259879
69. 10 Advanced Spatial Techniques – Geospatial Data Science ... https://warin.ca/geospatial/10-spatial_techniques.html
70. Advancements in Remote Sensing Applications https://isgindia.org/wp-content/uploads/2018/07/ISG-NL-JUNE-2018_FIN.pdf
71. 2.2.1 Simple feature geometry (sfg) <https://tmieno2.github.io/R-as-GIS-for-Economists/simple-feature-geometry-simple-feature-geometry-list-column-and-simple-feature.html>
72. Projections https://geopandas.org/en/stable/docs/user_guide/projections.html
73. Coordinate Reference Systems (CRS) <https://kodu.ut.ee/~kmoch/geopython2021/L2/crs-projections.html>
74. Map projections — Intro to Python GIS CSC documentation <https://automating-gis-processes.github.io/CSC/notebooks/L2/projections.html>
75. Map projections <https://autogis-site.readthedocs.io/en/2021/notebooks/L2/02-projections.html>
76. Working with map projections <https://pythongis.org/part2/chapter-06/nb/03-coordinate-reference-system.html>

77. Projection with transformation returns unexpected results ... <https://gis.stackexchange.com/questions/482445/projection-with-transformation-returns-unexpected-results-using-geopandas-compar>
78. Demo: GeoPandas and CRS https://uwgda-jupyterbook.readthedocs.io/en/latest/modules/04_Vector1_Geopandas_CRS_Proj/04_Vector1_Geopandas_CRS_Proj_demo.html
79. Hotspot and trend analysis of forest fires and its relation ... - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC9375094/>
80. Thermal Comfort Design Zone Delineation for India Using ... https://www.researchgate.net/publication/272568658_Thermal_Comfort_Design_Zone_Delineation_for_India_Using_GIS
81. mapping and detection of hotspot sources from industrial ... https://www.researchgate.net/publication/368386235_MAPPING_AND_DETECTION_OF_HOTSPOT_SOURCES_FROM_INDUSTRIAL_AREA_HEAT_IAH_USING_AERIAL_AND_SATELLITE-BASED_TIR_DATA_IN_PASIR_GUDANG
82. Surface Temperature Mapping to Assess Local Climate ... https://www.researchgate.net/publication/365746950_Surface_Temperature_Mapping_to_Assess_Local_Climate_Zones
83. Recognition of Thermal Hot and Cold Spots in Urban Areas ... <https://www.mdpi.com/2225-1154/6/1/16>
84. Spatial Characteristics and Temporal Trend of Urban Heat ... <https://www.mdpi.com/2076-3417/13/24/13323>
85. Hotspot and NDVI Differencing Synergy (HANDS) A New ... https://www.researchgate.net/publication/222543614_Hotspot_and_NDVI_Differencing_Synergy_HANDS_A_New_Technique_for_Burned_Area_Mapping_over_Boreal_Forest
86. Identified urban heat islands in India and the location ... https://www.researchgate.net/figure/Identified-urban-heat-islands-in-India-and-the-location-considered-for-analysis_fig9_348593212
87. Hotspot Analysis of Structure Fires in Urban Agglomeration <https://pdfs.semanticscholar.org/794b/85de92d4b46ab1af145c2169256a4c9906c3.pdf>
88. A Methodological Approach to Identify Thermal Anomaly ... <https://www.mdpi.com/2673-4931/29/1/47>
89. BallTree — scikit-learn 1.9.0 documentation <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.BallTree.html>
90. shapely.Polygon — Shapely 2.1.2 documentation <https://shapely.readthedocs.io/en/stable/reference/shapely.Polygon.html>

91. shapely/benchmarks/benchmarks.py at main <https://github.com/shapely/shapely/blob/main/benchmarks/benchmarks.py>
92. Difference between SDO_NN and BallTree Sklearn for ... <https://gis.stackexchange.com/questions/421092/difference-between-sdo-nn-and-balltree-sklearn-for-nearest-neighbors>
93. Benchmarking Nearest Neighbor Searches in Python <https://jakevdp.github.io/blog/2013/04/29/benchmarking-nearest-neighbor-searches-in-python/>
94. Acceleration for the Nearest Neighbor Search on Earth's ... <https://medium.com/aspectum/acceleration-for-the-nearest-neighbor-search-on-earths-surface-using-python-513fc75984aa>
95. How can I use Shapely to detect all points that are closer ... <https://stackoverflow.com/questions/55356929/how-can-i-use-shapely-to-detect-all-points-that-are-closer-than-n-meters>
96. Just Transition and Equitable Climate Action Resource ... <https://www.wri.org/just-transitions>
97. About Eutrophication and Hypoxia <https://www.wri.org/initiatives/eutrophication-and-hypoxia/learn>
98. Setting the Record Straight About Renewable Energy <https://www.wri.org/insights/setting-record-straight-about-renewable-energy>
99. 3 Ways to Manage Skyrocketing US Electricity Demand <https://www.wri.org/insights/managing-electricity-demand-growth-us>
100. Urban Heat Island Mapping with GIS: How Indian Cities ... <https://www.esri.in/en-in/newsroom/blog/gis-for-urban-heat-island-mapping>
101. <https://www.osti.gov/biblio/2572460> <https://www.osti.gov/biblio/2572460>:
102. An Artificial Intelligence Dataset for Solar Energy Locations in ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC9381711/>
103. IMPACT <https://mapyourgrid.org/impact/>
104. Thermal Summer Diurnal Hot-Spot Analysis: The Role of ... <https://www.mdpi.com/2072-4292/13/3/538>
105. Spatiotemporal Dynamics of Urban Heat Islands in India https://www.researchgate.net/publication/394651759_Spatiotemporal_Dynamics_of_Urban_Heat_Islands_in_India_A_Multivariate_Analysis_Using_High-Resolution_Remote_Sensing_Machine_Learning_and_Geospatial_Modeling_for_Climate-Resilient_Urban_Planning
106. A Dataset of the Operating Station Heat Rate for 806 Indian ... <https://arxiv.org/html/2410.00016v1>

107. India: Solar Atlas <https://hub.arcgis.com/maps/1f8f481732d048b0b4384eb586796113>
108. geopandas.sjoin_nearest https://geopandas.org/en/stable/docs/reference/api/geopandas.sjoin_nearest.html
109. Spatial join benchmarks (from Scipy 2020 talk) · Issue #114 <https://github.com/geopandas/dask-geopandas/issues/114>
110. geopandas spatial join extremely slow - python <https://gis.stackexchange.com/questions/175228/geopandas-spatial-join-extremely-slow>
111. STRtree — Shapely 2.1.1 documentation - Read the Docs <https://shapely.readthedocs.io/en/latest/strtree.html>
112. Scaling up spatial nearest joins : r/dataengineering https://www.reddit.com/r/dataengineering/comments/1cw2oru/scaling_up_spatial_nearest_joins/
113. The Map of Geospatial Data Science: How to Actually Learn It <https://milanjanosov.substack.com/p/the-map-of-geospatial-data-science>
114. Fast overlap finding for geographic datasets using GeoPandas <http://www.stevекlosterman.com/fast-overlap/>
115. Wrangling and joining 130M points with DuckDB + the open ... <https://dewey.dunnington.ca/post/2024/wrangling-and-joining-130m-points-with-duckdb--the-open-source-spatial-stack/>
116. How to extract CRS ellipsoid info in correct format for ... <https://gis.stackexchange.com/questions/366661/how-to-extract-crs-ellipsoid-info-in-correct-format-for-pyproj-inverse>
117. ENH: Geodesic Area, Length, and Distance Calculation ... <https://github.com/geopandas/geopandas/issues/3539>
118. Geometry Projection Speed Test - python <https://gis.stackexchange.com/questions/66130/geometry-projection-speed-test>
119. 10 GeoPandas/Shapely Moves for Serious Geospatial Work <https://medium.com/@ThinkingLoop/10-geopandas-shapely-moves-for-serious-geospatial-work-5ae1cd9523d8>
120. Pyproj (Transformers Groups) versus GeoPandas <https://gis.stackexchange.com/questions/418583/analysis-between-coordinate-transformation-libraries-pyproj-transformers-group>
121. Optimized Hot Spot Analysis (Spatial Statistics) - ArcGIS Pro <https://pro.arcgis.com/en/pro-app/3.5/tool-reference/spatial-statistics/optimized-hot-spot-analysis.htm>
122. Hot Spot Analysis (Getis-Ord Gi*) (Spatial Statistics) - ArcGIS Pro <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/spatial-statistics/hot-spot-analysis.htm>

123. What is Hotspot Analysis? - Geospatiality <https://glenbambrick.com/2016/01/21/what-is-hotspot-analysis/>
124. Hot Spot Spatial Analysis <https://www.publichealth.columbia.edu/research/population-health-methods/hot-spot-spatial-analysis>
125. Hotspot Analysis <https://www.nv5geospatialsoftware.com/docs/HotspotAnalysis.html>
126. Hot Spot Analysis Comparison (Spatial Statistics) - ArcGIS Pro <https://pro.arcgis.com/en/pro-app/3.5/tool-reference/spatial-statistics/hot-spot-comparison.htm>
127. How to calculate spatial hotspots and which tools do you ... <https://carto.com/blog/spatial-hotspot-tools/>
128. Mapping burnt areas using very high-resolution imagery and ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC12266457/>
129. Direct Thermal Imaging of Domain Wall Hot Spots in LiNbO₃ <https://pmc.ncbi.nlm.nih.gov/articles/PMC12757977/>
130. Normalized Hotspot Index (NHI): an original algorithm for ... https://www.asi.it/wp-content/uploads/2025/06/Pergola_CNR-IMAA.pdf
131. Solar Panel Hotspot Detection from Thermal Imagery Using ... https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7020619
132. Multivariate Geospatial Heat Map Analysis for ... <http://ui.adsabs.harvard.edu/abs/2025JAGG...13c...1R/abstract>
133. (PDF) Comparative Study of Urban Thermal Landscape ... https://www.researchgate.net/publication/399471201_Comparative_Study_of_Urban_Thermal_Landscape_Identification_Methods_Based_on_Landsat_Imagery
134. A Case Study of Very High and Medium-Resolution Data <https://www.mdpi.com/1424-8220/25/10/3009>
135. Assessment of Outdoor Thermal Comfort during the Last ... <https://www.mdpi.com/2673-4931/29/1/37>
136. Free Solar Maps & GIS Data | High-Res Download <https://solargis.com/resources/free-maps-and-gis-data?locality=india>
137. (PDF) An Artificial Intelligence Dataset for Solar Energy ... https://www.researchgate.net/publication/362719875_An_Artificial_Intelligence_Dataset_for_Solar_Energy_Locations_in_India
138. Global Dataset of Solar Power Plants <https://arxiv.org/pdf/2603.20601>
139. Historical data on photovoltaic (PV) energy generation in ... <https://ieee-dataport.org/documents/historical-data-photovoltaic-pv-energy-generation-india>
140. Solar Potential Estimation by Reconstructing 3D Model of ... <http://article.sapub.org/10.5923.j.ajgis.20200903.01.html>

141. Standard, Specification & Benchmark Cost <https://mnre.gov.in/en/solar-standard-specification-benchmark-cost/>
142. Hotspots of solar potential in India - WGBIS,CES,IISc https://wgbis.ces.iisc.ac.in/energy/paper/hotspots_solar_potential/RSER_hotspots.pdf
143. Project large collection of shapely object to different CRS <https://gis.stackexchange.com/questions/346181/project-large-collection-of-shapely-object-to-different-crs>
144. About the usability of distances and CRS · Issue #942 <https://github.com/geopandas/geopandas/issues/942>
145. Working with geographic coordinates in shapely? <https://gis.stackexchange.com/questions/260580/working-with-geographic-coordinates-in-shapely>
146. NATIONAL REMOTE SENSING CENTRE REPORT / ... <https://bhuvan-app3.nrsc.gov.in/data/download/tools/document/cm4.pdf>
147. Satellite (VIIRS) Thermal Hotspots and Fire Activity - ArcGIS Hub https://hub.arcgis.com/datasets/dece90af1a0242dcbf0ca36d30276aa3_0/about
148. How can I apply Hot spot analysis for Landsat calculated ... https://www.researchgate.net/post/How_can_I_apply_Hot_spot_analysis_for_Landsat_calculated_LST_values_in_ArcGIS
149. Geo Physical Products https://www.nrsc.gov.in/nrscnew/Dataproducs_geophysical_overview.php
150. Technical Methodology for Countrywide DEM and Ortho ... <https://bhuvan-app3.nrsc.gov.in/data/download/tools/document/SISDP-DEM-GENERATION-BHUVAN.pdf>
151. Anomaly Detection for Hotspot Identification in Landsat ... https://hal.science/hal-05233859v1/file/IGARSS24_Anomaly_Detection_Landsat.pdf
152. Persistent Hot Spot Detection and Characterisation Using ... https://www.researchgate.net/publication/326418833_Persistent_Hot_Spot_Detection_and_Characterisation_Using_SLSTR
153. NRSC Open EO Data Archive | NOEDA | Ortho | DEM | Elevation <https://bhuvan-app3.nrsc.gov.in/data/download/index.php?c=p&s=NICES&p=ff&g=TS>
154. India: Satellite VIIRS Thermal Hotspots and Fire Activity <https://www.arcgis.com/home/item.html?id=cb63469d22584493b5ab9a0f89f480ed>
155. Find coordinate of the closest point on polygon in Shapely <https://stackoverflow.com/questions/33311616/find-coordinate-of-the-closest-point-on-polygon-in-shapely>
156. The Shapely User Manual — Shapely 2.1.1 documentation <https://shapely.readthedocs.io/en/2.1.1/manual.html>

157. Nearest neighbour analysis <https://pythongis.org/part2/chapter-06/nb/07-nearest-neighbour.html>
158. Find nearest polygon (from GeoSeries) to point (from GeoSeries) <https://gis.stackexchange.com/questions/363891/find-nearest-polygon-from-geoseries-to-point-from-geoseries>
159. Nearest Neighbor Analysis for Large Datasets in Helsinki Region <https://towardsdatascience.com/nearest-neighbor-analysis-for-large-datasets-in-helsinki-79ebc7288c91/>
160. GitHub - erikbern/ann-benchmarks <https://github.com/erikbern/ann-benchmarks>
161. PiPNN: Ultra-Scalable Graph-Based Nearest Neighbor Indexing <https://arxiv.org/html/2602.21247v1>
162. Speeding up large-scale point-in-polygon test based spatial join on ... https://www.researchgate.net/publication/262274094_Speeding_up_large-scale_point-in-polygon_test_based_spatial_join_on_GPUs
163. Spatio-temporal changes pattern in the hotspot's footprint https://www.researchgate.net/publication/360603109_Spatio-temporal_changes_pattern_in_the_hotspot's_footprint_a_case_study_of_confirmed_recovered_and_deceased_cases_of_Covid-19_in_India
164. Top 10 Shapely Operations in Python (with Real OpenStreetMap Data) <https://www.youtube.com/watch?v=OutYkbflQ38>
165. STRtree — Shapely 2.1.2 documentation - Read the Docs <https://shapely.readthedocs.io/en/2.1.2/strtree.html>
166. Class 24: Spatial Data, OSMNX, GeoPandas https://asmithh.github.io/network-science-data-book/class_24_spatial.html
167. 第26章：与第三方库集成 - 博客园 <https://www.cnblogs.com/znlgis/p/20714690>
168. 7 GeoPandas/Shapely Tricks for Real-World Geospatial Analytics https://medium.com/@Nexumo_/7-geopandas-shapely-tricks-for-real-world-geospatial-analytics-931bad1cb79f
169. 3. Understanding Spatial Indexes https://understandingg.is/Spatial_Index/
170. Geospatial Analysis of Solar Energy in Riyadh Using a GIS ... <https://www.mdpi.com/2220-9964/10/5/291>
171. Mapping Solar Energy Potential of Southern India through ... https://www.researchgate.net/publication/326275091_Mapping_Solar_Energy_Potential_of_Southern_India_through_Geospatial_technology
172. Spatial analysis of solar parks in India <https://ideas.repec.org/p/osf/socarx/4nurv.html>

173. Analysis of Multi-renewable Energy Potential Sites in India ... <https://ui.adsabs.harvard.edu/abs/2024PIOS....8.1493S/abstract>
174. Solar power potential mapping in India using remote ... https://wgbis.ces.iisc.ernet.in/biodiversity/sahyadri_enews/newsletter/issue45/bibliography/Solar.pdf
175. How to speed up geopandas.sjoin_nearest using spatial ... <https://gis.stackexchange.com/questions/458745/how-to-speed-up-geopandas-sjoin-nearest-using-spatial-indexing>
176. STRtree — Shapely 2.1.2 documentation - Read the Docs <https://shapely.readthedocs.io/en/stable/strtree.html>
177. Question on Geopandas Spatial Join nearest <https://stackoverflow.com/questions/72639523/question-on-geopandas-spatial-join-nearest>
178. 4 Geometry operations - Geocomputation with Python <https://py.geocompx.org/04-geometry-operations>
179. Nearest (spatial) join as a new feature to geopandas? #1096 <https://github.com/geopandas/geopandas/issues/1096>
180. 10 GeoPandas/Shapely Moves for Real-World Maps & ... https://medium.com/@Nexumo_/10-geopandas-shapely-moves-for-real-world-maps-routing-ec4655c6615a
181. CRS to use for computing distances between points on ... <https://gis.stackexchange.com/questions/437826/crs-to-use-for-computing-distances-between-points-on-global-scale-with-geopandas>
182. Using GeoPandas to calculate distances across continents <https://gis.stackexchange.com/questions/457754/using-geopandas-to-calculate-distances-across-continents>
183. Nearest neighbor between point layer and line layer? [closed] <https://gis.stackexchange.com/questions/396/nearest-neighbor-between-point-layer-and-line-layer>
184. Calculating distance to nearest neighboring cell's edge, not centroid <https://forum.image.sc/t/calculating-distance-to-nearest-neighboring-cells-edge-not-centroid/78075>
185. New metric for BallTree - closest euclidean distance between two lines <https://github.com/scikit-learn/scikit-learn/issues/18260>
186. Proximity Analysis - Buffers, Nearest Neighbor - PyGIS https://pygis.io/docs/e_buffer_neighbors.html
187. Arham Ansari's UHI analysis workflow automated in Python https://www.linkedin.com/posts/mbforr_came-across-this-project-today-and-it-stopped-activity-7361592405130678272-n33G

188. Utilising geospatial data to understand urban heat island ... <https://meetingorganizer.copernicus.org/EGU26/EGU26-488.html?pdf>
189. India: Satellite (MODIS) Thermal Hotspots and Fire Activity <https://hub.arcgis.com/maps/a226e769bf3640d3b33275019ed53251>
190. Heat Vulnerability Index for Delhi: A Geospatial Analysis ... <https://ui.adsabs.harvard.edu/abs/2023AGUFMGC31E1091P/abstract>
191. Emerging heat stress patterns across India under future ... https://gfzpublic.gfz.de/rest/items/item_5037709_2/component/file_5038624/content
192. Ecosystem <https://geopandas.org/en/latest/community/ecosystem.html>
193. How can I accelerate a geopandas spatial join? <https://stackoverflow.com/questions/54804073/how-can-i-accelerate-a-geopandas-spatial-join>
194. Python: how to speed up spatial search (nearest point)? <https://gis.stackexchange.com/questions/73152/python-how-to-speed-up-spatial-search-nearest-point>
195. Applications | Bhuvan | NRSC Open EO Data Archive | NOEDA <https://bhuvan-app3.nrsc.gov.in/data/>
196. Bhuvan | NRSC Open EO Data Archive | NOEDA | Ortho | DEM <https://bhuvan-app3.nrsc.gov.in/data/download/index.php?c=p&s=NICES&g=OS&p=ocm3oc4>
197. ISRO's Geoportal | Gateway to Indian Earth Observation <https://bhuvan-app3.nrsc.gov.in/data/download/index.php?c=p&s=NICES&g=OS&p=ssh>
198. Satellite (VIIRS) Thermal Hotspots and Fire Activity <https://nationaldataplatform.org/catalog/dataset/satellite-viirs-thermal-hotspots-and-fire-activity>
199. Policy https://www.nrsc.gov.in/nrscnew/data_products_policy.php
200. Bhuvan | NRSC Open EO Data Archive | NOEDA | Ortho | DEM <https://bhuvan-app3.nrsc.gov.in/data/download/index.php?c=t&s=LV&p=vf>
201. Fast Point-in-Polygon Analysis with GeoPandas and Uber's ... <https://spatialthoughts.com/2020/07/01/point-in-polygon-h3-geopandas/>
202. CRS Compatibility Guide for Geospatial Python - pyproj 3.7 ... https://pyproj4.github.io/pyproj/stable/crs_compatibility.html
203. Identifying temperature 'hotspots' for increasing urban ... https://www.researchgate.net/publication/398898455_Identifying_temperature_'hotspots'_for_increasing_urban_resilience_to_heat_stress_in_Kolkata_West_Bengal_India
204. (PDF) Assessing urban heat island hotspots through virtual ... https://www.researchgate.net/publication/397911459_Assessing_urban_heat_island_hotspots_through_virtual_and_physical_boundary_analysis_with_space_syntax

205. Identification of urban hotspot areas from remote sensing ... https://www.researchgate.net/publication/398585629_Identification_of_urban_hotspot_areas_from_remote_sensing_towards_implementing_heat_mitigation_strategies_a_case_study_over_Navi_Mumbai_India
206. Hotspots and Coldspots: The Energy Wasting Culprits <https://www.akcp.com/2025/07/25/identifying-hotspots-through-temperature-monitoring/>
207. Satellite (MODIS) Thermal Hotspots and Fire Activity <https://hub.arcgis.com/maps/b8f4033069f141729ffb298b7418b653>
208. Satellite (VIIRS) Thermal Hotspots and Fire Activity <https://atlas.eia.gov/datasets/esri2::satellite-viirs-thermal-hotspots-and-fire-activity/about>
209. Satellite (VIIRS) Thermal Hotspots and Fire Activity - Datasets <https://wifire-data.sdsc.edu/dataset/satellite-viirs-thermal-hotspots-and-fire-activity>
210. Active Fire Data Attributes for MODIS and VIIRS <https://www.earthdata.nasa.gov/data/tools/firms/active-fire-data-attributes-modis-viirs>
211. Data Products: Fire/Hot Spot Characterization <https://www.goes-r.gov/products/baseline-fire-hot-spot.html>
212. Fire and Hot Spot Characterization https://www.star.nesdis.noaa.gov/goesr/product_land_fire.php
213. Datasets tagged hotspot in Earth Engine <https://developers.google.com/earth-engine/datasets/tags/hotspot>
214. Active Fire Monitoring with NASA FIRMS: Real-Time ... <https://offnadir-delta.com/blog/nasa-firms-active-fire-monitoring>
215. Land Surface Temperature – Products <https://ororatech.com/all-products/land-surface-temperature>
216. Ecosystem <https://geopandas.org/en/v0.14.2/community/ecosystem.html>
217. Nearest neighbor analysis with large datasets https://autogis-site.readthedocs.io/en/2021/notebooks/L3/06_nearest-neighbor-faster.html
218. Fast GeoSpatial Analysis in Python <https://matthewrocklin.com/blog/work/2017/09/21/accelerating-geopandas-1>
219. Managing Projections — GeoPandas 0.8.2 documentation <https://geopandas.org/en/v0.8.2/projections.html>
220. shapely.Point — Shapely 2.1.2 documentation <https://shapely.readthedocs.io/en/stable/reference/shapely.Point.html>
221. Algorithm for point to polygon distance calculations in ... <https://gis.stackexchange.com/questions/480964/algorithm-for-point-to-polygon-distance-calculations-in-shapely>

222. Challenge and Readiness to Implemented Geothermal ... <https://www.igminresearch.com/articles/html/igmin178>
223. a case study of Kamojang geothermal field in Garut, West ... https://www.researchgate.net/publication/369398883_Thermal_modeling_and_simulation_of_a_single-flash_geothermal_power_plant_involving_non-condensable_gas_a_case_study_of_Kamojang_geothermal_field_in_Garut_West_Java_Indonesia
224. Case study: Rethinking geothermal energy in Indonesia <https://re-course.org/newsupdates/rethinking-geothermal-energy-in-indonesia/>
225. Decarbonisation of Thermal Power Plants and CCS ... <https://www.eria.org/research/decarbonisation-of-thermal-power-plants-and-ccs-business-feasibility-study-in-indonesia>
226. Hotspot Analysis of Structure Fires in Urban Agglomeration <https://www.mdpi.com/2571-6255/4/3/38>
227. Loading... <https://arcticdata.io/catalog/view/doi%3A10.18739%2FA2N87311N>
228. 1463800 - ESS-DIVE <https://data.ess-dive.lbl.gov/view/doi:10.15485/1463800>
229. Tropical Forest Firefighters Trained with Space-Based ... <https://ororatech.com/resources/news-blog/tropical-forest-firefighters-trained-with-space-based-wildfire-detection-system-to-combat-deforestation>
230. ITDAV-25 (Indian Thermal Dataset for Autonomous Vehicles) <https://iee-dataport.org/documents/itdav-25-indian-thermal-dataset-autonomous-vehicles>
231. <https://ieeexplore-ieee-org.unr.idm.oclc.org/docum...> <https://ieeexplore-ieee-org.unr.idm.oclc.org/document/10059640>:
232. Towards Real-Time Monitoring of Thermal Peaks in ... <https://www.mdpi.com/1424-8220/22/15/5904>
233. IoT System Based on Artificial Intelligence for Hot Spot ... <https://www.mdpi.com/1424-8220/23/15/6749>
234. Human Detection in Aerial Thermal Images Using Faster R ... <https://www.mdpi.com/2079-9292/11/7/1151>
235. Hotspot Detection in Photovoltaic Array Using Thermal ... https://www.researchgate.net/publication/359086238_Hotspot_Detection_in_Photovoltaic_Array_Using_Thermal_Imaging_Method
236. Study and Analysis of Thermal Imaging Sensors for Object ... <https://www.internationaljournalsr.org/IJECE/paper-details?Id=440>
237. Thermal imaging-based identification of facial features in ... <https://iaesjournal.org/index.php/ijict/article/download/2861/2644>

238. (PDF) Performance Comparison of Several Deep Learning ... https://www.researchgate.net/publication/357472657_Performance_Comparison_of_Several_Deep_Learning-Based_Object_Detection_Algorithms_Utilizing_Thermal_Images
239. Range Performance Modelling of Thermal Imaging System ... https://www.researchgate.net/publication/327838448_Range_Performance_Modelling_of_Thermal_Imaging_System_based_on_Single_Parameter_Characterised_by_Ambient_Temperature_and_Relative_Humidity
240. Bhuvan | NRSC Open EO Data Archive | NOEDA | Ortho | DEM <https://bhuvan-app3.nrsc.gov.in/data/download/index.php?c=p&s=NICES&p=nppm&g=TS>
241. Nearest neighbor analysis with large datasets <https://autogis-site.readthedocs.io/en/2019/notebooks/L3/nearest-neighbor-faster.html>
242. How can I introduce the radius in query radius-BallTree ... <https://stackoverflow.com/questions/63121268/how-can-i-introduce-the-radius-in-query-radius-balltree-sklearn-with-units-of-r>
243. Using Scikit-learn's Binary Trees to Efficiently Find Latitude ... <https://towardsdatascience.com/using-scikit-learns-binary-trees-to-efficiently-find-latitude-and-longitude-neighbors-909979bd929b/>
244. Finding an Observation's Nearest Neighbors in Python ... <https://blog.stackademic.com/finding-an-observations-nearest-neighbors-in-python-using-scikit-learn-5b0aadd48dec>
245. Shapely Distance different from Geopy/Haversine <https://gis.stackexchange.com/questions/184340/shapely-distance-different-from-geopy-haversine>
246. Benchmarking osm and osm-edge data planes <https://dev.to/flomesh/benchmarking-osm-and-osm-edge-data-planes-3c65>
247. OSM+: Billion-Level Open Street Map Dataset for City-wide ... <https://arxiv.org/html/2512.06743v1>
248. Efficient Structures for Movement Data in Python <https://pdfs.semanticscholar.org/f7b6/eeae7759924ebcc938de25f756dba2809928.pdf>
249. Reduce memory-footprint of point-in-polygon lookups ... <https://gis.stackexchange.com/questions/495673/reduce-memory-footprint-of-point-in-polygon-lookups-in-geopandas>
250. GeoPandas performance: optimizing vectorized operations <https://github.com/geopandas/geopandas/issues/430>
251. geopandas.sjoin <https://geopandas.org/en/stable/docs/reference/api/geopandas.sjoin.html>
252. Spatial Join with GeoPandas (and GEOS) <https://aetperf.github.io/2021/09/10/Spatial-Join-with-GeoPandas-and-GEOS.html>

253. STRTree.query that returns index instead of geometry object <https://github.com/Toblerity/Shapely/issues/615>
254. Shapely Implementation Dramatically Slower than PostGIS <https://gis.stackexchange.com/questions/218214/shapely-implementation-dramatically-slower-than-postgis>
255. python - Shapely STRtree way too slow? <https://stackoverflow.com/questions/61025297/shapely-strtree-way-too-slow>
256. How to Efficiently use Shapely in Python for spatial ... <https://medium.com/@lukelashley99/how-to-efficiently-use-shapely-in-python-for-spatial-operations-such-as-intersections-b07edd725f56>
257. performance comparison of spatial indexing structures for https://mavmatrix.uta.edu/context/cse_theses/article/1337/type/native/viewcontent
258. Benchmarking and Engineering Data Structures for ... <https://arxiv.org/pdf/2607.07367>
259. Performance benchmark on mdds R-tree <https://kohei.us/2019/02/15/performance-benchmark-on-mdds-rtree/>
260. How could I speed up my written python code: spheres ... <https://stackoverflow.com/questions/71104627/how-could-i-speed-up-my-written-python-code-spheres-contact-detection-collisio>
261. BallTree query match time is $O(n)$ not $O(\log(n))$ #19066 <https://github.com/scikit-learn/scikit-learn/issues/19066>
262. query_ball_point — SciPy v1.18.0 Manual https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.KDTree.query_ball_point.html
263. Solved: Best spatial references for models using distance <https://community.esri.com/t5/arcgis-pro-questions/best-spatial-references-for-models-using-distance/td-p/1157786>
264. Choosing projection / CRS for short-distance based ... <https://gis.stackexchange.com/questions/148181/choosing-projection-crs-for-short-distance-based-analysis>
265. Measurement accuracy for small distances - geodesic vs. ... <https://gis.stackexchange.com/questions/208162/measurement-accuracy-for-small-distances-geodesic-vs-planar-utm-wgs84>
266. Geodesic or planar: which to use for distance analysis <https://www.esri.com/arcgis-blog/products/arcgis-pro/analytics/geodesic-or-planar-which-to-use-for-distance-analysis>
267. Choose a CRS for Distance Measurement - Bathyl <https://www.bathyl.com/en/blog/how-to-choose-a-crs-for-distance-measurement>

268. Projected vs Geographic Coordinate Reference Systems <https://earthdatascience.org/courses/earth-analytics/spatial-data-r/geographic-vs-projected-coordinate-reference-systems-UTM/>
269. A Quick Note on Coordinate Systems and Map Projections https://isaacullah.github.io/GIS_Workshops/02-A_Quick_Note_on_Projections.pdf
270. ISRO's Geoportal | Gateway to Indian Earth Observation <https://bhuvan-app3.nrsc.gov.in/data/download/index.php?c=p&s=NICES&g=CS&p=smf>
271. Identification, Characterization and Correction of ... <https://www.recentadvancesin.com/remote-sensing/3020-6448-rars25008/>
272. A Guide to H3, S2, and Discrete Global Grid Systems <https://forrest.nyc/discrete-global-grid-systems-h3-s2-vs-lat-long/>
273. Chapter 1 Introduction | Spatial Data and Analyses in R <https://wec.wur.nl/r/spatial/introduction.html>